

Strategy Improvement and Empirical Results

5398 Assignment 1



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0. Introduction

The original portfolio construction method in this project was based on a naïve equal-weight allocation, where each asset in the portfolio was assigned the same weight regardless of its return characteristics or risk profile. While simple and easy to implement, this approach does not account for differences in expected returns or volatility across assets, which can lead to suboptimal performance.

1. Strategy Specification

To improve upon this baseline, a **momentum- and risk-based allocation strategy** was introduced. Specifically, assets were ranked using a combination of **6-month (126-day) and 12-month (252-day) momentum signals**, while skipping the most recent 21 trading days to avoid short-term reversal effects. From this ranking, the top **7 assets (TOP_K = 7)** were selected at each rebalancing step. This replaces the original “hold everything equally” logic with a selective, signal-driven portfolio.

```
mom6 = past.iloc[-1] / past.iloc[-LOOKBACK_MOM_6] - 1.0  
mom12 = past.iloc[-1] / past.iloc[-LOOKBACK_MOM_12] - 1.0  
  
score = (0.5 * mom6 + 0.5 * mom12).dropna()
```

In addition to improving asset selection, the weighting scheme was fundamentally enhanced. Instead of equal weights, the new strategy applies **inverse-volatility weighting** using a 63-day lookback window. This ensures that more volatile assets receive lower allocations, thereby improving risk balance across the portfolio. A **maximum weight constraint of 30% per asset** was also imposed to prevent excessive concentration.

```
vol = past[top].tail(LOOKBACK_VOL).std(axis=0).replace(0.0, np.nan)  
inv_vol = (1.0 / vol).replace([np.inf, -np.inf], np.nan).fillna(0.0)
```

Transaction costs were incorporated by calculating daily portfolio turnover as the absolute change in asset weights and applying a fee rate of 0.1%. This adjustment reduces gross portfolio returns by the estimated trading cost, making the backtest more realistic than a frictionless equal-weight strategy.



```
gross_ret = (w * r).sum(axis=1)  
turnover = w.diff().abs().sum(axis=1)  
fee_rate = 0.001  
cost = fee_rate * turnover  
cost.iloc[0] = 0.0
```

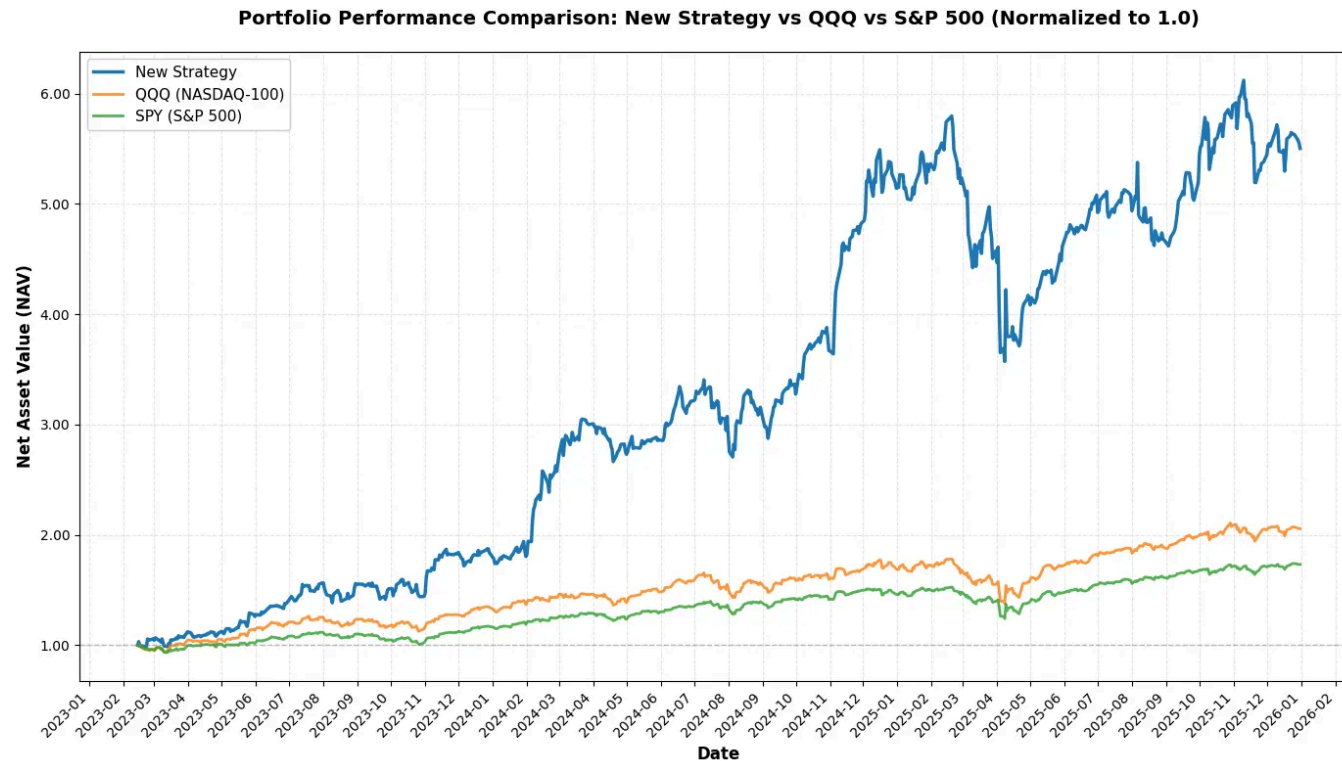
2. Results & Visualization

The empirical results demonstrate that the proposed improvements significantly altered the portfolio's performance characteristics relative to the original equal-weight strategy. The enhanced strategy achieved an **annualized return of 83.66%**, substantially outperforming both Invesco QQQ Trust (28.60%) and SPDR S&P 500 ETF Trust (21.13%). This large increase in return indicates that the introduction of momentum-based selection and top-K filtering was highly effective in capturing strong-performing assets and amplifying portfolio growth, which is also clearly visible in the NAV curve where the strategy grows from 1.0 to above 5.5 over the sample period, far exceeding the relatively flat trajectory of the original equal-weight approach.

	Annual Return	Annual Vol	Sharpe	Max Drawdown
New Strategy	0.836616	0.390681	1.751850	-0.405455
QQQ	0.285984	0.197577	1.371813	-0.227683
SPY	0.211325	0.152966	1.329725	-0.187552

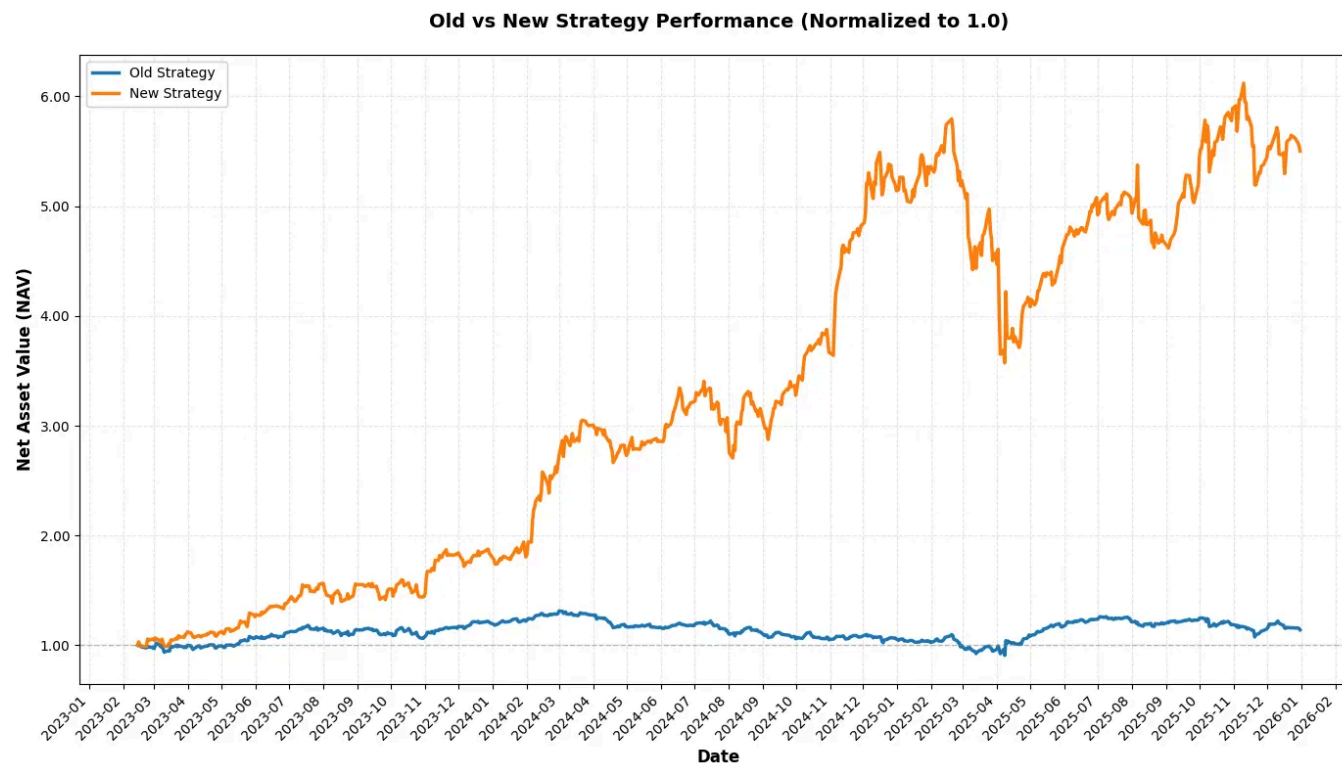
In terms of risk, the strategy exhibits a higher **annualized volatility of 39.07%**, compared to 19.76% for QQQ and 15.30% for SPY. This increase in volatility reflects the more aggressive and concentrated nature of the

strategy, as it allocates capital to a smaller set of high-momentum stocks. However, despite this higher risk, the strategy achieves a **Sharpe ratio of 1.75**, outperforming both QQQ (1.37) and SPY (1.33). This indicates that the strategy delivers **superior risk-adjusted returns**, meaning that the additional volatility is compensated by disproportionately higher returns.



The **maximum drawdown of -40.55%** is notably larger than that of QQQ (-22.77%) and SPY (-18.76%), highlighting a key trade-off. While the strategy

generates significantly higher returns, it is also more exposed to downside risk during adverse market conditions. This is consistent with the behavior observed in the NAV plot, where sharp declines occur during periods of market reversal, followed by relatively strong recoveries. These fluctuations reflect the inherent sensitivity of momentum strategies to changes in market trends.



3. Limitations

Despite the strong performance, several limitations remain in the current implementation. First, the strategy is evaluated on a **single continuous dataset without a proper train–test split**, meaning there is no clear separation between in-sample and out-of-sample performance. As a result, the reported returns may be **overfitted to the specific time period** and may not generalize well to unseen market conditions. Second, the parameter choices (e.g., lookback windows, TOP_K = 7, and weight cap) are fixed and not validated through systematic tuning or cross-validation. Third, although transaction costs are included, the model does not account for **market impact, slippage, or liquidity constraints**, which could materially reduce real-world performance. Finally, the strategy relies heavily on momentum signals, making it vulnerable to **sudden market reversals and regime shifts**, as reflected in its relatively large drawdowns.

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